MACHINE LEARNING

Heart failure risk prediction

A classification model built to miss as few high-risk patients as possible

The problem

299
patients in the dataset, 90 held out for testing

12
clinical features such as age, ejection fraction, and serum creatinine

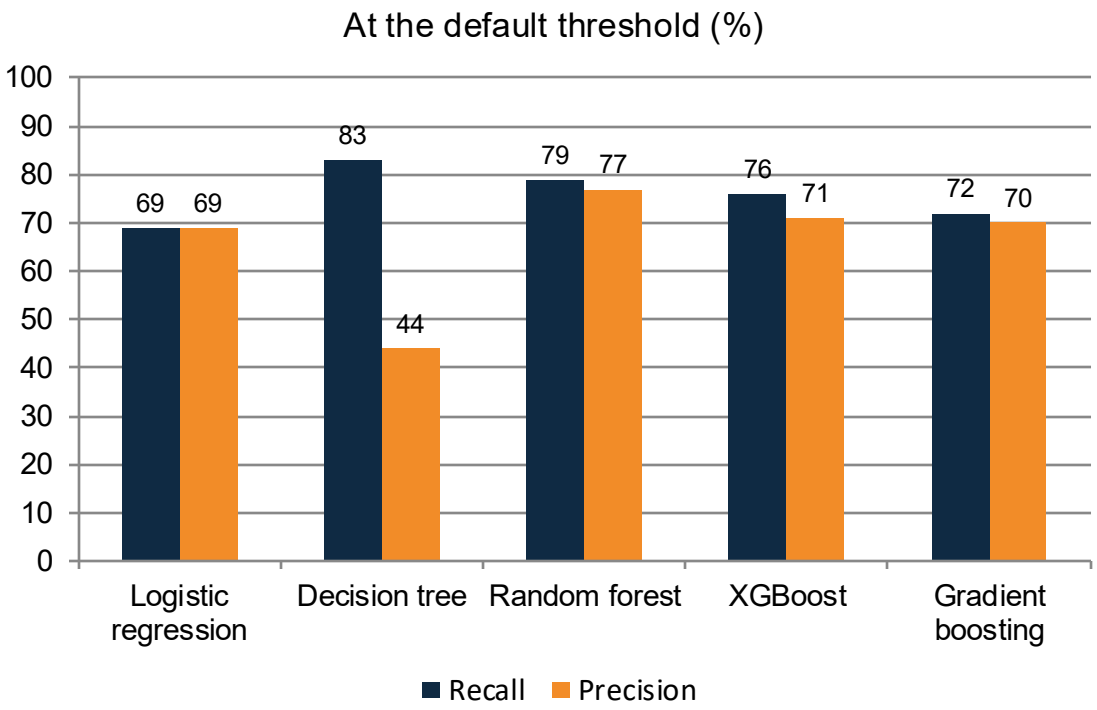
32%
of patients died, so the classes are imbalanced

- The goal is to predict whether a heart failure patient will die or survive
- Missing a high-risk patient costs far more than a false alarm, so recall is the metric that matters most

How it was built

1. Stratified 70/30 train and test split, so both sets keep the same share of deaths
2. SMOTE on the training data only, to balance the classes without leaking into the test set
3. Five models: logistic regression, decision tree, random forest, XGBoost, gradient boosting
4. Tuned the decision threshold to reach at least 90% recall, then chose the best of the models that got there
5. Explained predictions with SHAP and audited fairness by age group and gender

Results: recall and precision by model



93.1%

recall for XGBoost after tuning the threshold

56.2%

precision at that point, the best of the high-recall models

Catching more patients means more false alarms. That trade-off was chosen on purpose for a clinical use.

What drives risk, and who the model serves

Top features (SHAP importance)

- Follow-up time
- Ejection fraction
- Serum creatinine
- Creatinine phosphokinase
- Platelets

Fairness audit

16.7
point recall gap across age groups

9.1
point recall gap between men and women

Both gaps pass the 5-point alert line, so the audit flags bias to fix before any use.

Before this could be used

- Re-test without follow-up time. It is the top feature, but it would not be known when a prediction is made
- Work with more data: 299 patients and a 90-patient test set give wide uncertainty
- Close the age and gender recall gaps found by the fairness audit
- The notebook includes demo versions of drift monitoring, hospital record integration, and a shadow-mode trial. They are blueprints, not live systems
- The notebook’s own readiness checklist reads "not fully ready"